

Mid-level Backend Developer Technical Assessment

Candidate Name:

Date:

Phone:

Email:

Please read this instruction carefully before you start.

- This entire test takes 120 minutes
- Each answer is graded on depth, completeness and for codes, readability, maintainability and efficiency.
- No Googling, no ChatGPT or any other AIs.
- Please share your video and entire screen throughout the test. No external help.
- You can use your favorite IDE to code and copy your code snippets in text (not images) to this document.
- In each section, there is a time recommendation. Please try to stick to the time recommendation for each section as much as possible so that you can score a higher overall grade.
- There are 3 sections in total - you should read through the questions and do the ones you are most comfortable with first.
- English only. All the answers should be written in this document. No external links or attachments.
- Everything about the questions is contained herein. Please do not ask our invigilators about the questions.

Good luck!

Practice Test for Middle Backend Developer Position

Part 1: NodeJS

1.1. Write a JavaScript function that takes a string as input and returns the count of each unique word in the string. Ignore punctuation and consider case-insensitive matching. For example, given the input "The cat and the dog were playing in the garden.", the function should return `{ the: 2, cat: 1, and: 1, dog: 1, were: 1, playing: 1, in: 1, garden: 1 }`.

1.2. Describe how you would handle user authentication and authorization in a NodeJS application. What tools or libraries would you use, and what security considerations would you take into account?

1.3. Explain the concept of middleware in Express.js and provide an example of how you would use middleware in a NodeJS application.

1.4. Imagine you're creating a RESTful API with NodeJS. Please provide an example of a PUT endpoint that would update user data based on an ID. Assume that you're using Express.js and a PostgreSQL database.

Part 2: Crypto Trading & Uniswap

2.1. Explain the concept of decentralized exchanges (DEX) and how they differ from traditional centralized exchanges. Discuss the advantages and disadvantages of using a DEX like Uniswap.

2.2. In Uniswap, what is the purpose of liquidity pools and how do they work? Describe the role of liquidity providers and the benefits they receive for participating in the Uniswap ecosystem.

2.3. Describe the process of adding liquidity to a Uniswap pool. Include the steps involved, the risks associated with providing liquidity, and any considerations that liquidity providers should keep in mind.

Bonus: Algorithms and Efficiency

3.1. Implement a sorting algorithm of your choice in any programming language you prefer. Explain the time complexity of the algorithm and discuss its advantages and disadvantages.

3.2. Describe a situation where you had to optimize the performance of a backend application. What techniques or strategies did you employ, and what were the results?